

How to use the scripts (step by step)

Project must be **Unity 6 + URP**.

Copy Scripts/ into Assets/MyWorld/Scripts/ first and wait for compile.

Also create layers/tags:

Tags: Player

Layers (recommended): Ground, Vehicle, Water, Interactable

A) PLAYER

Scripts

- PlayerMotor.cs — movement
- PlayerInteraction.cs — Press **E** interact
- PlayerSpawnPoint.cs — spawn marker
- PlayerBootstrap.cs — snap to spawn on Play
- SimpleFollowCamera.cs — temporary camera

Setup player (click path)

1. Hierarchy → **3D Object** → **Capsule** (or drop a character model)
2. Rename Player
3. Set Tag = **Player**
4. Add component **Character Controller**
5. Add component **Player Motor**
6. Add component **Player Interaction**
7. Create empty child CameraTarget at head height (optional)
8. Hierarchy → **Camera** (or use Main Camera)
9. Add **Simple Follow Camera**
10. Drag Player into Target
11. Create empty PlayerSpawn → add **Player Spawn Point** → place on terrain
12. Create empty Systems → add **Player Bootstrap** → assign Player Motor

Controls

- **WASD** move
- **Left Shift** run
- **Space** jump
- **Mouse** look (follow camera)
- **E** interact

Download if you want a better character

- Asset Store: Starter Assets Third Person (optional; our PlayerMotor works without it)
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B) HOUSE / DOOR

Scripts

- HouseDoor.cs
- BuildingInteriorVolume.cs

Setup enterable house

1. Place a house prefab in the scene
2. Create empty child Door_Interact
3. Add a **Box Collider** on it (not required to be trigger)
4. Add **House Door**
5. Create empty InteriorSpawn inside house floor
6. Create empty ExteriorSpawn outside door
7. Drag both into House Door fields
8. Mode = **Teleport To Interior**
9. Optional: add **Building Interior Volume** with trigger box covering interior

Setup swinging door only

1. Put House Door on door mesh pivot
2. Mode = **Toggle Door Rotation**
3. Assign Door Pivot
4. Press E to open/close

Assets to download for houses

- Fantasy / medieval village modular pack (URP)
 - Medieval props (barrels, chairs) for interiors
 - Optional tavern interior pack
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C) CARS

Scripts

- CarController.cs
- VehicleSeat.cs
- VehicleEnterExit.cs

Setup car (important — do carefully)

1. Drop car model in scene
2. Root object: add **Rigidbody**
 - Mass: 1200–1600
 - Interpolation: Interpolate
 - Collision Detection: Continuous
3. Create empty child COM slightly below center → assign as Center Of Mass
4. Create 4 children with **Wheel Collider**:
 - WC_FL, WC_FR, WC_RL, WC_RR
 - Place at wheel positions
5. Add **Car Controller** on root → assign 4 WheelColliders + optional visual wheels
6. Add **Vehicle Seat** on root (or seat empty at driver position)
7. Add **Vehicle Enter Exit** on same object as seat/root
8. Create empty ExitPoint beside car → assign on Vehicle Seat
9. Layer = Vehicle (optional)

Play

- Walk near car, look at it, press **E** to enter
- **WASD** drive
- **E** exit

Assets

- Included possible models in AssetsToImport (Jeep, etc.) if copied
 - Or download: driveable car URP, wheel collider car
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D) BIKES

Scripts

- BikeController.cs
- VehicleSeat.cs
- VehicleEnterExit.cs

Setup bike

1. Bike model + **Rigidbody** mass 200–280
2. Two WheelColliders: front + rear
3. Add **Bike Controller** → assign wheels
4. Add **Vehicle Seat** + **Vehicle Enter Exit**
5. Exit point beside bike

Controls

Same enter/exit as car. Lean is automatic while turning.

Assets

- Motorcycle / bike model pack (URP materials)
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E) PLANES

Scripts

- PlaneController.cs
- VehicleSeat.cs
- VehicleEnterExit.cs

Setup plane

1. Plane model + **Rigidbody**
2. Add **Plane Controller**
3. Add seat + enter/exit
4. Place on flat runway / beach strip

Controls

- **W** thrust
- **A/D** yaw
- **Q/E** roll
- **R/F** pitch
- **E** exit

Assets to download

- Simple airplane model (URP)
 - Optional runway / wind sock props
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F) BOATS

Scripts

- BoatController.cs
- - seat / enter-exit

Setup boat

1. Boat model + Rigidbody
2. Add **Boat Controller**

3. Set Water Level Y to your ocean/lake height
4. Add seat + enter/exit

Assets

- Boat models (low poly boats OK)
 - Pier models
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G) WORLD HELPERS

ZoneTrigger

1. Empty object with Box Collider (Is Trigger)
2. Add **Zone Trigger**
3. Set names (Zone_Beach, display "Beach")
4. Scale box to cover zone

DayNightCycle

1. Empty DayNight
2. Add **Day Night Cycle**
3. Assign Directional Light (sun)
4. Adjust Day Length Minutes

AmbientZoneAudio

1. Empty with trigger collider + Audio Source
2. Assign looping ambient clip (waves/wind/birds)
3. Add **Ambient Zone Audio**

Assets to download for audio/sky

- Skybox pack
 - Nature ambience audio (waves, wind, forest, village)
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H) Interaction system note

Anything interactable should implement IInteractable

Ready-made:

- HouseDoor
- VehicleEnterExit

Player must have PlayerInteraction.

Temporary on-screen prompt uses OnGUI (fine for setup).
Replace later with uGUI / UI Toolkit for production UI.

I) Common mistakes

Problem	Fix
Cannot enter car	Need PlayerInteraction + VehicleEnterExit + collider on vehicle
Car explodes / flips	Lower center of mass; check WheelCollider positions; reduce torque
Bike falls over	Keep COM low; test on flat ground first
Plane won't lift	Hold W for speed; pitch with R/F
Boat sinks / flies	Adjust Water Level Y and buoyancy
Fall through ground	CharacterController + TerrainCollider; Ground mask
Scripts missing	Reimport Scripts folder; wait compile
Pink model	URP/Lit materials

J) Professional order of use

1. Player walk solid
2. One house door
3. One car
4. Zones + audio
5. Day/night
6. Bike / boat / plane
7. Then decorate world using the big world-building guide